

ACTL3182 Wk4

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Factor Model

The return of some asset i can be explained by a factor,

$$\tilde{r}_i = \alpha_i + \beta_{i,1}f_1 + \epsilon_i,$$

where ϵ_i satisfies our usual error assumptions.

Consider an evenly weighted portfolio of n assets. Then,

$$\sigma_{P,n}^2 = \beta_{P,n}^2 \sigma_f^2 + \sum_{i=1}^n \frac{1}{n^2} \sigma_{\epsilon_i}^2.$$

We assume that

$$\lim_{n \rightarrow \infty} \sigma_{P,n}^2 = \beta_P^2 \sigma_f^2,$$

that is, idiosyncratic risk of a portfolio can be made arbitrarily small.

Deriving APT

Assume no arbitrage and say asset returns can be described by the single factor model,

$$\tilde{r}_i = a_i + b_{i,1}f_1 + \epsilon_i, \quad i = 1, 2, \dots, N.$$

WLOG, suppose that $\mathbb{E}[f_1] = 0$ and $\text{Var}(f_1) = 1$ (otherwise, we can make a new factor and then reparameterise our model), and so

$$\mathbb{E}[\tilde{r}_i] = a_i.$$

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Further, assume that the universe of stocks is large, and so we can construct portfolios with arbitrarily small idiosyncratic risk. So, from here we ignore idiosyncratic risk,

$$\tilde{r}_i = a_i + b_{i,1}f_1, \quad i = 1, 2, \dots, N.$$

Deriving APT (cont.)

Say we have two such portfolios,

$$\tilde{r}_i = a_i + b_{i,1}f_1,$$

$$\tilde{r}_j = a_j + b_{j,1}f_1.$$

Goal: Make a risk free portfolio by having 0 exposure to the factor.

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If w is the weight put on asset i , then we want

$$wb_{i,1} + (1 - w)b_{j,1} = 0 \implies w = \frac{b_{j,1}}{b_{j,1} - b_{i,1}}$$

and hence

$$\tilde{r}_P = w\tilde{r}_i + (1 - w)\tilde{r}_j = \frac{a_i b_{j,1}}{b_{j,1} - b_{i,1}} + \frac{a_j b_{i,1}}{b_{i,1} - b_{j,1}}$$

is a riskless portfolio.

Deriving APT (cont.)

By no arbitrage, if there is a risk free asset with return r_f , we must have

$$r_P = r_f.$$

So, let λ_0 be the risk free rate, and so

$$\lambda_0 = \frac{a_i b_{j,1}}{b_{j,1} - b_{i,1}} + \frac{a_j b_{i,1}}{b_{i,1} - b_{j,1}}.$$

Rearranging this to get all i and j terms on each side,

$$\frac{a_i - \lambda_0}{b_{i,1}} = \frac{a_j - \lambda_0}{b_{j,1}}.$$

Deriving APT (cont.)

But our choice of i and j were arbitrary, and so this ratio must be a constant, say λ_1 ,

$$\begin{aligned}\lambda_1 &:= \frac{a_i - \lambda_0}{b_{i,1}} \implies a_i = \lambda_0 + \lambda_1 b_{i,1} \\ &\implies \mathbb{E}[\tilde{r}_i] = \lambda_0 + \lambda_1 b_{i,1}.\end{aligned}$$

APT

In the k factor model, the expected returns of an asset is given by

$$r_i = \lambda_0 + \lambda_1 b_{i,1} + \cdots + \lambda_k b_{i,k},$$

where λ_0 is the risk free rate and $b_{i,j}$ is the exposure of the asset to the j th factor.